

MTH101 MIDTERM PREPARATION GRAND QUIZ SOLUTION LECTURE 1 TO 22 ALI KHAN



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MTH101

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MIDTERM PREPARATION

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ALI KHAN 0316-9948478

Question # 1 of 30 (Start time: 05:18:43 PM, 21 June 2021)

Which of the following is equation of a line that passes through the point (5, -3) and whose slope is -1?

Select the correct option

- | | |
|----------------------------------|--------------|
| ☒ | $y = -x + 2$ |
| ☐ | $y = -x - 2$ |
| ☐ | $y = x - 2$ |
| ☐ | $y = x + 2$ |

[Click to Save Answer](#)

Question # 2 of 30 (Start time: 05:19:35 PM, 21 June 2021)

Which of the following is VERTEX of parabola represented by the equation : $y = x^2 + 2$

Select the correct option

[Reload](#)

- | | |
|----------------------------------|---------------|
| ☐ | None of these |
| ☒ | (0, 2) |
| ☐ | (2, 6) |
| ☐ | (1, 3) |

[Click to Save Answer & Move to Next Question](#)

Question # 3 of 30 (Start time: 05:20:20 PM, 21 June 2021)

What is the derivative of $\sin(x^2) + \cos(2x)$?

Select the correct option

- | | |
|----------------------------------|----------------------------|
| ☐ | $2x \cos(x^2) + 2 \sin 2x$ |
| ☒ | $2x \cos(x^2) - 2 \sin 2x$ |
| ☐ | None of these. |
| ☐ | $2x \sin(x^2) + 2 \cos 2x$ |

Question # 4 of 30 (Start time: 05:20:47 PM, 21 June 2021)

If $y = f(g(x))$ then its derivative will be

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | $\frac{d}{dx} f(x) \cdot \frac{d}{dx} (g(x))$ |
| ☒ | $\frac{d}{dx} f(g(x)) \cdot \frac{d}{dx} (g(x))$ |

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Question # 5 of 30 (Start time: 05:21:15 PM, 21 June 2021)

In xy-plane both x and y are positive in _____ quadrant.

Select the correct option

☒	First
☐	Second
☐	Forth
☐	Third

Question # 6 of 30 (Start time: 05:21:45 PM, 21 June 2021)

If $f(x) = 2x$ and $g(x) = x^2 + 3x$, then the derivative of $f(x) \times g(x)$ is

Select the correct option

☐	$x^2 + 12.$
☐	$12x^2 + 6x.$
☒	$6x^2 + 12x.$
☐	None of these.

[Click to Save Answer &](#)

Question # 7 of 30 (Start time: 05:22:18 PM, 21 June 2021)

What is the derivative of $\cos ec(x^4)$?

Select the correct option

☐	$\cos ec(x^4) \cot(x^4)$
☐	$-\cos ec(x^4) \cot(x^4)$
☐	None of these
☒	$-4x^3 \cos ec(x^4) \cot(x^4)$

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Question # 8 of 30 (Start time: 05:22:54 PM, 21 June 2021)

Which of the following rule will be used to find the derivative of $xcotx$?

Select the correct option

 Reload

☐	sum
☒	product
☐	power
☐	difference

Question # 8 of 30 (Start time: 05:22:54 PM, 21 June 2021)

Which of the following rule will be used to find the derivative of $xcotx$?

Select the correct option



☐	sum
☒	product
☐	power
☐	difference

Question # 9 of 30 (Start time: 05:24:02 PM, 21 June 2021)

Let $f(x) = x - 3$ and $g(x) = x^3$ then $(f - g)(x) = ?$

Select the correct option



- | | |
|----------------------------------|----------------|
| ☐ | $-x^3 - x - 3$ |
| ☐ | $x^3 + x - 3$ |
| ☐ | $-x^3 + x + 3$ |
| ☒ | $-x^3 + x - 3$ |

Click to Save Answer & Move to Next Question

Question # 10 of 30 (Start time: 05:24:32 PM, 21 June 2021)

Derivative of $\sin(u^5)$ is.....

Select the correct option



- | | |
|----------------------------------|------------------|
| ☒ | $5u^4 \cos u^5$ |
| ☐ | $-5u^4 \cos u^5$ |
| ☐ | $5u^4 \sin u^5$ |
| ☐ | $-5u^4 \sin u^5$ |

Click to Save Answer & Move to Next Question

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Question # 11 of 30 (Start time: 05:25:01 PM, 21 June 2021)

Which of the following describes the equation:
 $x = 1$?

Select the correct option

- | | |
|----------------------------------|---------------------------|
| ☐ | A point (1, 0) |
| ☐ | A line parallel to x-axis |
| ☒ | A line parallel to y-axis |
| ☐ | A point (0, 1) |

Question # 12 of 30 (Start time: 05:25:34 PM, 21 June 2021)

If $f(x) = x \sin x^2$, then which of the following is true about it

Select the correct option



- | | |
|----------------------------------|--|
| ☐ | <i>Its derivative w.r. t x is $2x \cos x^2 + \sin x^2$</i> |
| ☒ | <i>Its derivative w.r. t x is $2x^2 \cos x^2 + \sin x^2$</i> |
| ☐ | <i>None of these</i> |
| ☐ | <i>Its derivative w.r. t x is $x \cos x^2 + \sin x^2$</i> |

[Click to Save Answer & Mark](#)

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Question # 13 of 30 (Start time: 05:26:01 PM, 21 June 2021)

The derivative of composition of functions can be found by using _____.

Select the correct option

☐	Power rule
☐	Generlized derivative formula
☐	None of these
☒	Chain rule

Click to

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Question # 14 of 30 (Start time: 05:26:32 PM, 21 June 2021)

For the curve $f(t) = -23t + 25$, the slope of the tangent line at point 't = -1' is _____.

Select the correct option

☐	23
☐	-25
☐	25
☒	-23

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Question # 15 of 30 (Start time: 05:26:58 PM, 21 June 2021)

The derivative of x^3+x^2 is _____.

Select the correct option

- | | |
|----------------------------------|-----------|
| ☐ | $3x-2$ |
| ☒ | $3x^2+2x$ |
| ☐ | $3x+2$ |
| ☐ | $3x^2-2x$ |

Question # 16 of 30 (Start time: 05:27:28 PM, 21 June 2021)

If $f(x) = \cos x^5$, then which of the following is true about it

Select the correct option



- | | |
|----------------------------------|--|
| ☐ | None of these. |
| ☒ | Its derivative w. r. t x is $-5x^4 \sin x^5$ |
| ☐ | Its derivative w. r. t x is $5x^4 \sin x^5$ |
| ☐ | Its derivative w. r. t x is $-x^5 \sin x^4$ |

[Click to Save Answer & Move to Next Question](#)

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Question # 17 of 30 (Start time: 05:27:54 PM, 21 June 2021)

The interval $(-3, 2)$ denote the set of all _____ between -3 and 2.

Select the correct option

☐	Integers
☐	Natural numbers
☐	Rational numbers
☒	Real numbers

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Question # 18 of 30 (Start time: 05:28:21 PM, 21 June 2021)

A function $f(x) = x / |x|$ is

Select the correct option



not continuous at $x = 1$



continuous at $x = 1$

Question # 19 of 30 (Start time: 05:28:50 PM, 21 June 2021)

Derivative of a graph of straight line bisecting the first and third quadrants is-----

Select the correct option

☐	Infinity
☒	+1
☐	-1
☐	Zero

Click to Save A

Question # 20 of 30 (Start time: 05:29:23 PM, 21 June 2021)

Which of the following is solution of an inequality:
 $|x + 3| > 2$

Select the correct option

- | | |
|----------------------------------|----------------------|
| ☒ | $x < -5$ or $x > -1$ |
| ☐ | $x > -5$ or $x < -1$ |
| ☐ | $x < -5$ or $x < -1$ |
| ☐ | $x > -5$ or $x > -1$ |

Click to S

Question # 21 of 30 (Start time: 05:29:48 PM, 21 June 2021)

Which of the following is not an example of set?

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | Collection of prime numbers which are even. |
| ☒ | Collection of some stars in the sky. |
| ☐ | Collection of whole numbers between 20 and 25. |
| ☐ | Collection of colors in rainbow. |

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Question # 22 of 30 (Start time: 05:30:16 PM, 21 June 2021)

If a function f has a relative extrema at a point c , then c is a critical point for f .

Select the correct option

☒	True
☐	False

cf

Question # 23 of 30 (Start time: 05:30:45 PM, 21 June 2021)

The interval $[a, b)$ is defined by

Select the correct option

- | | |
|----------------------------------|---------------------------|
| ☒ | $\{x : a \leq x < b\}$ |
| ☐ | $\{x : a < x < b\}$ |
| ☐ | $\{x : a \leq x \leq b\}$ |
| ☐ | $\{x : a < x \leq b\}$ |

Click to S

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Question # 24 of 30 (Start time: 05:31:09 PM, 21 June 2021)

$$|-7| + |-4| = \underline{\hspace{2cm}}$$

Select the correct option

- | | |
|----------------------------------|----------------|
| ☐ | -11 |
| ☐ | Does not exist |
| ☒ | 11 |
| ☐ | ± 11 |

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Question # 25 of 30 (Start time: 05:31:49 PM, 21 June 2021)

Which of the following is midpoint of the line segment joining points (1, 3) and (1, 5)?

Select the correct option

- | | |
|----------------------------------|---------|
| ☐ | (0, 2) |
| ☒ | (1, 4) |
| ☐ | (2, 8) |
| ☐ | (0, -1) |

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Question # 26 of 30 (Start time: 05:32:24 PM, 21 June 2021)

The derivative of a variable with respect to itself is equal to

Select the correct option

☐	2
☐	8
☐	0
☒	1

Question # 27 of 30 (Start time: 05:32:56 PM, 21 June 2021)

If $g(X) = 3x^2$ then $g'(2)$ will be ?

Select the correct option

- | | |
|----------------------------------|-----|
| ☐ | 18 |
| ☐ | -12 |
| ☐ | 15 |
| ☒ | 12 |

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Question # 28 of 30 (Start time: 05:33:26 PM, 21 June 2021)

Let a function f is twice differentiable at a stationary point x_0 ,
then f has relative maximum at x_0 if

Select the correct option

- | | |
|----------------------------------|-------------------|
| ☒ | $f''(x_0) < 0$ |
| ☐ | $f''(x_0) = 0$ |
| ☐ | $f''(x_0) > 0$ |
| ☐ | $f''(x_0) \geq 0$ |

Question # 29 of 30 (Start time: 05:33:50 PM, 21 June 2021)

The derivative of $(x^2) + 2\sin x$ w. r. t x is

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | $-2x-2\cos x$ |
| ☐ | $2x-2\cos x$ |
| ☐ | $-2x+2\sin x$ |
| ☒ | $2x+2\cos x$ |

Click

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Question # 30 of 30 (Start time: 05:34:24 PM, 21 June 2021)

The derivative of $(1/\sin x)$ is

Select the correct option

- | | |
|----------------------------------|---|
| ☒ | $-\operatorname{Cosec} x \cdot \cot(x)$ |
| ☐ | $\operatorname{Cosec} x \cdot \cot(x)$ |
| ☐ | $\sec(x)$ |
| ☐ | $\operatorname{Cosec}(x)$ |

Question # 8 of 30 (Start time: 05:55:47 PM, 21 June 2021)

What is the derivative of $\sin(x^\pi)$?

Select the correct option

- | | |
|----------------------------------|-------------------------------|
| ☐ | $-\pi(x)^{\pi-1} \cos(x^\pi)$ |
| ☒ | $\pi(x)^{\pi-1} \cos(x^\pi)$ |
| ☐ | $\cos(x^\pi)$ |
| ☐ | None of these |

Click to S

Question # 9 of 30 (Start time: 05:56:16 PM, 21 June 2021)

Which of the following is VERTEX of parabola represented by the equation : $y = x^2 + 2$

Select the correct option

[Reload N](#)

- | | |
|----------------------------------|---------------|
| ☐ | (1, 3) |
| ☐ | (2, 6) |
| ☒ | (0, 2) |
| ☐ | None of these |

[Click to Save Answer & Move to N](#)

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Question # 10 of 30 (Start time: 05:56:52 PM, 21 June 2021)

$$d(5x+4)/dx=.....$$

Select the correct option

- | | |
|----------------------------------|---|
| ☐ | 2 |
| ☐ | 9 |
| ☐ | 4 |
| ☒ | 5 |

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Question # 11 of 30 (Start time: 05:57:19 PM, 21 June 2021)

$2x + 3y = 6$ is

Select the correct option

- | | |
|----------------------------------|----------------|
| ☒ | A function |
| ☐ | Not a function |

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Question # 12 of 30 (Start time: 05:57:43 PM, 21 June 2021)

The implicit differentiation of $y^2 = x$, gives $\frac{dy}{dx} = \underline{\hspace{2cm}}$.

Select the correct option

- | | |
|----------------------------------|--------------------|
| ☒ | $\frac{1}{2y}$ |
| ☐ | 1 |
| ☐ | $2y \frac{dy}{dx}$ |
| ☐ | $\frac{1}{2x}$ |

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Question # 13 of 30 (Start time: 05:58:06 PM, 21 June 2021)

If $xy=3$, then $dy/dx=-3/x$

Select the correct option

☒	False
☐	True

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Question # 14 of 30 (Start time: 05:58:40 PM, 21 June 2021)

Which of the following is distance between the points (2, 1) and (1, 1)?

Select the correct option

☒	1
☐	Square root of (13)
☐	(1, 0)
☐	-1

Question # 15 of 30 (Start time: 05:59:10 PM, 21 June 2021)

Derivative of $\sin(5x + 3)$ is

Select the correct option

 Report

☒	$5\cos(5x + 3)$
☐	$-5\sin(5x + 3)$
☐	$3\cos(5x + 3)$
☐	$3\cos 5x$

[Click to Save Answer & Move](#)

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Question # 16 of 30 (Start time: 05:59:48 PM, 21 June 2021)

Which of the following describes the equation:
 $x = 1$?

Select the correct option

- | | |
|----------------------------------|---------------------------|
| ☐ | A point (1, 0) |
| ☒ | A line parallel to y-axis |
| ☐ | A point (0, 1) |
| ☐ | A line parallel to x-axis |

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Question # 17 of 30 (Start time: 06:00:39 PM, 21 June 2021)

Which of the following is distance between the points (1, 2) and (1, 3)?

Select the correct option

☒	1
☐	(0, -1)
☐	(2, 5)
☐	0

Question # 18 of 30 (Start time: 06:01:01 PM, 21 June 2021)

If $f(x) = x^5$ and $g(x) = x^2$, then the derivative of $\frac{f(x)}{g(x)}$ is

Select the correct option

☐	3x
☒	$3x^2$
☐	None of these
☐	$\frac{3}{x^2}$

[Click to Save Answer](#)

Question # 19 of 30 (Start time: 06:01:28 PM, 21 June 2021)

The derivative of $\sec(3x)$ is

Select the correct option

- | | |
|----------------------------------|---------------------|
| ☐ | $3\sec 3x \tan x$ |
| ☐ | $\sec 3x \tan 3x$ |
| ☒ | $3 \sec 3x \tan 3x$ |
| ☐ | $3 \sec x \tan x$ |

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Question # 20 of 30 (Start time: 06:01:57 PM, 21 June 2021)

If for any positive number ϵ (epsilon) we can find δ (delta) such that $| (3x-5) - 1 | < \epsilon$,if x satisfies $0 < |x-2| < \delta$
Then $f(x) = \dots\dots\dots$

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | None of these |
| ☒ | $3x-5$ |
| ☐ | $x-2$ |
| ☐ | $3x-5- 1$ |

Question # 21 of 30 (Start time: 06:02:29 PM, 21 June 2021)

Which of the following rule will be used to find the derivative of $xcotx$?

Select the correct option

☐	difference
☐	sum
☒	product
☐	power

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Question # 22 of 30 (Start time: 06:02:59 PM, 21 June 2021)

$$\lim \left[\frac{f(x)}{g(x)} \right] = \frac{\lim f(x)}{\lim g(x)}; \quad \lim g(x) \text{ is}$$

Select the correct option

- | | |
|----------------------------------|-------------------|
| ☐ | Equal to zero |
| ☒ | Not equal to zero |
| ☐ | One |
| ☐ | Undefined |

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Question # 23 of 30 (**Start time: 06:03:21 PM, 21 June 2021**)

Velocity is the rate of change of position w.r.t

Select the correct option

☒	time
☐	force
☐	acceleration
☐	displacement

Question # 24 of 30 (Start time: 06:03:48 PM, 21 June 2021)

If $f(x) = \cos x^5$, then which of the following is true about it

Select the correct option

☒	<i>Its derivative w.r.t x is $-5x^4 \sin x^5$</i>
☐	<i>Its derivative w.r.t x is $5x^4 \sin x^5$</i>
☐	None of these.
☐	<i>Its derivative w.r.t x is $-x^5 \sin x^4$</i>

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Question # 25 of 30 (Start time: 06:04:11 PM, 21 June 2021)

If $x - y$ is positive, then

Select the correct option

- | | |
|----------------------------------|---------|
| ☐ | $y > x$ |
| ☐ | $x = y$ |
| ☒ | $x > y$ |
| ☐ | $x < y$ |

Question # 26 of 30 (Start time: 06:04:39 PM, 21 June 2021)

The function f is said to be _____ on (a, b) , if $f''(x) > 0$ on that interval.

Select the correct option

☐	Concave down
☐	Constant
☒	Concave up
☐	None of these

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Question # 27 of 30 (Start time: 06:05:08 PM, 21 June 2021)

A functionbe differentiable at the point of discontinuity.

Select the correct option

☒	cannot
☐	can

Question # 28 of 30 (Start time: 06:05:31 PM, 21 June 2021)

At a corner a tangent line does not exist because the slopes of the secant lines do not have a (two sided) limit.

Select the correct option

☒	True
☐	False

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Question # 29 of 30 (Start time: 06:05:54 PM, 21 June 2021)

Which of the following point satisfies the equation: $y = -3x + 2$

Select the correct option

- | | |
|----------------------------------|-----------|
| ☐ | $(-1, 4)$ |
| ☐ | $(-3, 2)$ |
| ☐ | $(1, 4)$ |
| ☒ | $(2, -4)$ |

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Question # 30 of 30 (Start time: 06:06:25 PM, 21 June 2021)

The inequality $L - e < f(x) < L + e$ can be written as

Select the correct option



$$|f(x) - e| < L$$



$$|f(x) - L| > e$$



$$|f(x) - e| > L$$



$$|f(x) - L| < e$$

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